MILITARY SPECIFICATION

ADAPTER, TONE-SIGNALLING TA-977( )/PT

This specification is approved for use by CORADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers a tone-signalling device, designated as Adapter, Tone-Signalling TA-977( )/PT.

2. APPLICABLE DOCUMENTS.

2.1 Issues of Documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-handling, Wood
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications Research & Development Command
ATTN: DRDCO-PE-EC-3
Fort Monmouth, New Jersey 07703

by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

FSC 5805

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PPP-B-640	Box, Fiberboard, Corrugated, Triple Wall
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Rock
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment
MIL-M-13231	Marking for Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam)
MIL-P-26514	Polyurethane Foam

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

DRAWINGS

COMMUNICATIONS RESEARCH & DEVELOPMENT COMMAND

DL-SM-B-951620	Adapter, Tone-Signalling TA-977( )/PT

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish sample units for First Article inspection and approval (See 4.3).

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3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and Drawing and Data List DL-SM-B-951620, (including the requirements for parts, materials, and processes hereon) (See 4.4).

3.3 Parts, Materials and Processes in General. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (See 4.4).

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.5 Marking.

3.5.1 General. Marking shall conform to specification MIL-M-13231. (See 4.4).

3.5.2 Serial Numbers. The equipment shall be serially numbered.

3.6 Interchangeability. Like units, assemblies, sub-assemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454. (See 4.8).

3.7 Pre-conditioning. The equipment shall be capable of meeting all the requirements herein without subsequent processing after subjection to the bounce pre-conditioning of section 4.6 (mounted in the MX-2915/PT or the MX-230A/PT Accessory Case). (Also see section 4.5).

3.8 ELECTRICAL

3.8.1 Operating Current. The current drain from the battery shall not exceed 8.4 milliamperes at 70°F.

3.8.2 Non-Operating Current. Less than 1.0 microamperes shall be drawn from the battery unless a Keyset button is depressed.

3.8.3 Output Signals.

3.8.3.1 Output Level. The Dual-Tone Signal output level from the equipment into 600 ohms shall be -7 dBm ± 2 dB for each digit.

3.8.3.2 Output Frequencies. The equipment shall create the standard, 16 Dual-Tone Signals. The frequencies of the Dual-Tone Signals for each digit in Hertz (Hz ± 1.0%) are:

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	High Group			
Low Group	1209	1336	1477	1633
697	1	2	3	FO
770	4	5	6	F
852	7	8	9	I
941	R	0	C	P

3.8.3.3 Output Level vs Battery Voltage. The output level of the Dual-Tone Signal shall not vary more than 0.5 dB as the battery voltage decreases to 6.5 VDC and not more than 3 dB when the battery voltage decreases to 5.4 VDC.

3.8.3.4 Signal Rise-Time. The rise-time of any of the Dual-Tone Signals shall be no greater than 25 milli-seconds.

3.8.3.5 High-Frequency Emphasis. The level of the higher-frequency component of the Dual-Tone Signal shall be at least 0.5 dB but not more than 4.5 dB greater than the lower-frequency component.

3.8.4 Distortion. The total harmonic distortion of the output signal shall be less than 2.5%.

3.8.5 Protection. The circuitry of the equipment shall include protection against damage by excessive voltage surges and normal 20 Hz ring signals. A continuous 20 Hz ring signal of 90 VRMS impressed across the output terminals of an Adapter when de-activated shall cause a current greater than 25.0 milliamperes RMS to flow.

3.8.6 Isolation. The output circuitry of the equipment shall not allow more than 50 microamperes of direct current to flow through it (when connected to a 24 VDC power supply). It shall not be adversely affected by the polarity of switchboard or transmission line voltages.

3.8.7 Impedance. The output impedance of the equipment shall be 600 ohms $\pm$ 10% when activated and at least 40k ohms when deactivated.

3.8.8 Operation. The equipment shall generate the appropriate Dual-Tone Signals as each button of the operations keyset is depressed in accordance with 4.9.7.

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3.9 Environmental. The equipment shall meet the following environmental conditions.

3.9.1 Temperature.

3.9.1.1 Operating. The equipment shall be capable of operating in an ambient temperature range of -30°F (without solar radiation) to +150°F (including solar radiation).

3.9.1.2 Non-Operating. The equipment shall be capable of storage in an ambient temperature range of -70°F to +160°F.

3.9.2 Altitude. The equipment shall show no evidence of damage or deterioration in performance at altitudes up to 10,000 feet above sea level after storage at altitudes up to 50,000 feet above sea level.

3.9.3 Rain. The Adapter shall meet the operating requirements of 3.8.8 and shall sustain no physical damage, during periods of heavy precipitation with intermittent wind.

3.9.4 Humidity. During and after testing as specified in 4.10.4 the equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling, or deterioration of parts and materials, and shall meet the requirements of 3.8.

3.9.5 Fungus. After the fungus test of 4.10.5, the equipment shall be visually examined using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than .015 square inches in area, of sparse microbal growth as evidenced by growth colonization which includes branching and sporulation on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation (See 6.8).

3.9.6 Salt Fog. After the salt fog test of 4.10.6, the equipment when examined visually with aid of a low power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface, or exfoliation of metal (See 6.8.6).

3.9.7 Bounce Loose Cargo. Following testing as specified in 4.10.7, the equipment shall meet the requirements of 3.8.

3.9.8 Bench Handling. Following testing as specified in 4.10.8, the equipment shall meet the requirements of 3.8.

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3.10 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268 (See 4.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (See 4.3). Does not include preparation for delivery.

b. Inspections covered by subsidiary documents (See 4.4).

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of packaging (See 4.11).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish six (6) First Article units of Adapter, Tone-Signalling TA-977( )/PT.

4.3.2 First Article Inspection. The First Article inspection shall consist of the inspections specified in Table I, and shall be performed in the order specified therein.

4.3.3 First Article Data. The First Article test plan and test report(s) shall be as required in the contract or purchase order.

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4.3.4 Failures. One or more failures shall be cause for refusal to grant First Article approval.

TABLE I

First Article Inspection

Insp	Reqmt	Test	Units
Inspection covered by Subsidiary Documents	3.2 3.3 3.4 3.5	4.4	All Units
Preconditioning	3.7	4.6	All Units
Group A	Table II		All Units
Group B	Table III		All Units
Group C	Table IV		Units 1, 2, & 3

4.4 Inspections Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract or purchase order.

ITEM	WHERE REQUIRED
Construction	3.2
Parts, Materials & Processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality Conformance Inspection Before Preparation for Delivery. The contractor shall perform the inspections specified in 4.4, 4.6, and 4.5.1 through 4.5.3, in that order. This does not relieve the contractor of his responsibilities for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to verification sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

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4.5.1.1 Order of Inspection Within Group A. Group A inspection shall be performed in an order that is satisfactory to the Government.

4.5.2 Group B Inspection. Group B inspection shall normally be performed on lots that have passed Group A inspection on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B Sampling Plans. The Group B sampling plans for the AQL's listed in Table III and the inspection level of MIL-STD-105 shall be as follows:

AQL	Inspection Level
4.0 DPHU	S-4
6.5%	S-4

4.5.2.2 Order of Inspection Within Group B. Group B inspection shall be performed in an order that is satisfactory to the Government.

TABLE II

Group A Inspection

			AQL	
Insp	Reqmt	Insp	Major	Minor
Visual Mechanical	3.10	4.7	1%	4%
Electrical:	3.8	4.9		
Operating Current	3.8.1	4.9.1		
Non-Operating Current	3.8.2	4.9.1		
Output Level	3.8.3.1	4.9.2.1	1.0 DPHU for the group	*
Output Frequencies	3.8.3.2	4.9.2.2		
Output Level vs Battery Voltage	3.8.3.3	4.9.2.3		
Operation	3.8.8	4.9.7	.40%	

*All electrical and operational defects are considered major.

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TABLE III

Group B Inspection

Insp	Reqmt	Insp	AQL
Interchangeability	3.6	4.8	6.5%
Signal Rise-Time	3.8.3.4	4.9.2.4	4.0 DPHU for the group
High-Frequency Emphasis	3.8.3.5	4.9.2.5	
Distortion	3.8.4	4.9.3	
Protection	3.8.5	4.9.4	
Isolation	3.8.6	4.9.5	
Impedance	3.8.7	4.9.6	

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C Inspection. Three sample units shall be selected by the Government representative at random for all inspections in Table IV every 3 months of 1500 units or fraction thereof produced, whichever comes first. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in the order specified in Table IV.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A and Group B inspection.

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TABLE IV

Group C Inspection

Insp	Reqmt	Insp	Unit 1	2	3
Low Temperature	3.9.1.1	4.10.1.1	1		
High Temperature	3.9.1.2	4.10.1.2	2		
Altitude	3.9.2	4.10.2	3		
Rain	3.9.3	4.10.3	6		
Humidity	3.9.4	4.10.4		1	
Fungus	3.9.5	4.10.5			1
Salt Fog	3.9.6	4.10.6		2	
Bounce	3.9.7	4.10.7	5		
Shock:Bench Handling	3.9.8	4.10.8	4		

4.6 Bounce Preconditioning. The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, made by the L.A.B. Corporation, Skaneateles, N. Y, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection.

4.7 Visual and Mechanical Inspection of Equipment. The equipment shall be given a complete mechanical and visual inspection and test to determine that the quality of all materials and workmanship is in compliance with the requirements of 3.10 of this specification and for the defects listed in MIL-STD-252.

4.8 Inspection for Dimensional Interchangeability. The dimensions listed below shall be measured to determine compliance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

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d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.9 Electrical.

4.9.1 Current Drain. (Use Figure 1 for the Test Set-Up).

a. The unit shall be powered from its battery during the test.

b. Set the current meter (A) to a high scale - 100 milliamperes.

c. With none of the buttons of the Keyset depressed, step the current meter (A) to a more sensitive scale starting from the highest scale. The meter should read less than 1 microampere.

d. Return the current meter (A) to a 30 or 100 milliamperes scale.

e. Position switches; S1 to position 2 and S2 to position 1.

f. Press and hold down any digit key.

(1) The voltmeter (V3) should read at least 7.5 Vdc. If it does not, change the battery.

(2) The current meter (A) should read not more than 8.4 milliamperes at 70°F and not more than 10 milliamperes at the temperature limits specified in 3.9.1.1.

4.9.2 Output Signals. (Use Figure 1 for the Test Set-Up).

4.9.2.1 Output Level.

a. Place Switch S1 in position 2, and Switch S2 in position 1.

b. Press and hold down the number 1 digit key.

c. Check the Battery Voltage (meter V3) (with the digit key down). If not 7.5 Vdc or more replace and repeat steps b and c.

d. The AC VTVM (V2) shall read -7 dBm $\pm$ 2.0 dB.

e. Repeat steps b, c and d for each digit of the keyset.

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4.9.2.2 Output Frequencies.

a. Switches set as in 4.9.2.1.

b. Press and hold down the number 1 and the number 2 digit key simultaneously.

c. Check the Battery Voltage (meter V3). If not 7.5 Vdc or more replace it and repeat steps b and c.

d. The Frequency Counter (C) should read 697 Hz ± 1%.

e. Press and hold down both the number 1 and the number 4 digit keys simultaneously.

f. The Frequency Counter should read 1209 Hz ± 1%.

g. Repeat steps b through f for each digit key, first pressing an adjacent horizontal key simultaneously and then pressing an adjacent vertical key simultaneously. Refer to the chart in paragraph 3.8.3.2. When two horizontal keys are pressed simultaneously the common low group frequency should be obtained. When two vertical keys are pressed simultaneously the common high group frequency should be obtained.

4.9.2.3 Output Level vs Battery Voltage.

a. Switches set as in 4.9.2.1.

b. Substitute an adjustable laboratory power supply in place of the battery.

c. Set the power supply to 7.5 Vdc.

d. Record the AC VTVM (V2) reading (in decibels).

e. Reset the power supply to 6.5 Vdc.

f. Record the AC VTVM (V2) reading (in decibels).

g. The difference between the reading in step d and f should be no greater than 0.5 dB.

4.9.2.4 Signal Rise-Time. (Use Figure 1 for the Test Set-Up).

a. Set switch S2 in position 1 and S1 in position 2.

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b. Depress each button of the Keyset one at a time.

c. Observe on the Oscilloscope (S) that the rise time of the Dual-Tone Signals for each digit does not exceed 25 milliseconds.

4.9.2.5 High-Frequency Emphasis.

a. Repeat the steps of 4.9.2.2.

b. Record the level of the low group frequency and the level of the high group frequency for each digit.

c. The level difference between the low group frequency and the high group frequency for each digit must comply with 3.8.3.5.

4.9.3 Distortion.

a. Repeat the steps of 4.9.2.2.

b. Measure the distortion for each of the low group frequencies and for each of the high group frequencies. (On Distortion Analyser D).

c. The total harmonic distortion of each frequency shall not exceed 2.5%.

4.9.4 Protection. (Use Figure 2 for the Test Set-Up) Worst Case Test.

a. The battery must be connected during this test.

b. Set the AC VTVM to the 10 volt scale.

c. With no buttons of the keyset depressed, crank the G-42A/PT Ringing Generator.

d. An AC voltage greater than 2.5 volts AC should be observed.

4.9.5 Isolation. (Use Figure 3 for the Test Set-UP).

a. The battery must be connected during this test.

b. Set the Power Supply to 24 volts dc.

c. Set the current meter (A) to a high scale, 100 milliamperes.

d. With none of the buttons of the Keyset depressed step the current meter to a more sensitive scale starting from the highest scale. The meter should read less than .05 milliamperes direct-current.

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e. Repeat steps c and d with one of the buttons of the Keyset depressed. The current should be less than .05 milliamperes.

4.9.6 Impedance. (Use Figure 1 for the Test Set-Up).

a. Non-Operating Impedance of the Unit.

(1) Place switches S1 and S2 in position 1.

(2) Set the Oscillator at a frequency of 1 kHz.

(3) Set attenuator for 10 dB of attenuation.

(4) Adjust Oscillator level control so that AC VTVM, V1, reads 0 dBm.

(5) Place S2, in position 2 (leave S1 in position 1).

(6) AC VTVM V1, should read no less than -6.0 dBm.

b. Operating Impedance of the Unit.

(1) Place switches S1 and S2 in position 1.

(2) Press a button of the Keyset.

(3) Read AC VTVM, V2, on the dB scale.

(4) Place S1 in position 2 (leave S2 in position 1). Repeat steps (2) and (3). AC VTVM, V2, shall read between 5.5 dB and 6.5 dB less than the previous measurement.

4.9.7 Operation. Connect the Unit, to the Emergency Operator's terminals on the SB-22( )/PT Switchboard and interconnect from operator's position plug through a TA-222/PT Line Pack terminated by a 600 ohm resistor across its terminals. Depress each button of the Keyset and ascertain by listening, via the Operator's headset whether two frequencies are being generated. If there is any doubt about the signal, place an oscilloscope across the 600 ohm resistor and view the signal.

4.10 ENVIRONMENTAL CONDITIONS.

4.10.1 Temperature. During and after testing, the equipment shall meet the requirements of 3.8.

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4.10.1.1 Low Temperature. The equipment shall be subjected to the test of Method 502.1, Procedure 1 of MIL-STD-810. The storage temperature shall be maintained for a period of not less than 2 hours following stabilization of the test item.

4.10.1.2 High Temperature. The equipment shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810. For Step 7 the internal chamber temperature shall be adjusted to 150°F for test item operation (this includes effects of solar radiation).

4.10.2 Altitude Test. The equipment shall be subjected to the test of Method 500, Procedure 1, of MIL-STD-810. The operational test is the only electrical test required.

4.10.3 Rain Test. The equipment shall be subjected to the test of Method 506, Procedure 1 of MIL-STD-810. The equipment shall be in its operational configuration except that the SB-22 Switchboard shall be shielded from the rain in such a way that only the adapter is subjected to the test. The test shall be performed following bounce testing of the same sample test item.

4.10.4 Humidity Test. The equipment shall be subjected to the test of Method 507, Procedure II of MIL-STD-810. Measurements performed during the prescribed periods shall determine conformance to the requirements of 3.8.

4.10.5 Fungus. The equipment shall be subjected to the test of Method 508, Procedure 1 of MIL-STD-810. Gasket sealed or other nonhermetic sealed test items shall be opened during exposure and internal areas shall be sprayed with spore suspension as well as external surfaces.

4.10.6 Salt Fog. The equipment shall be subjected to the test of Method 509, Procedure 1 of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finishes on metal surfaces.

4.10.7 Bounce. The equipment shall be subjected to the test of Method 514.2, Procedure XI, Part 2 of MIL-STD-810.

4.10.8 Bench Handling. The equipment shall be subjected to the test of Method 516.2, Procedure V of MIL-STD-810.

4.11 Quality Conformance Inspection of Packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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4.12 Rough Handling Test (Packaging). When the rough handling test, in accordance with MIL-P-116, is required by the contract, the Operation test of 4.9.7 shall be conducted to determine freedom from malfunction caused by rough handling.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (See 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Adapter, Tone-Signalling TA-977( )/PT shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Adapter, Tone-Signalling TA-977( )/PT shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative Application. Not required.

5.1.1.4 Unit Packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical Literature. When furnished, technical literature shall be individually unit packed method IC-1.

5.1.1.4.2 Adapter, Tone-Signalling TA-977( )/PT. Each adapter shall be individually unit packed method III as follows: Cushion each adapter on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternative, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close-fitting fiberboard box conforming to PP-B-636, W5c. When furnished, place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specifications.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial Preservation. Preservation shall be in accordance with MIL-STD-1188.

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5.2 Packing. Packing shall be level A, or B or commercial, as specified (See 6.2b). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Adapter, Tone-Signalling TA-977( )/PT, unit packed as specified in 5.1 shall be packed within a close fitting box conforming to PPP-B-601, overseas type PPP-B-621, style 4 class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds or the container length and width is 48 X 24 inches or more and the weight exceeds 100 pounds, 3 X 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781,type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Adapter, Tone-Signalling TA-977( )/PT, unit packed as specified in 5.1, shall be packed within a close fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-60, respectively; selection of the material and application shall be in accordance with the appendix of applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 X 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Adapter, Tone-Signalling TA-977( )/PT, unit packed as specified in 5.1, shall be placed on a pallet,

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load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial Packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES.

6.1 Intended Use. Tone-Signalling Adapter TA-977( )/PT is a 4 X 4 button keyset with associated electronic circuitry which is used with the Switchboard SB-22 ( )/PT to provide a sixteen (16) digit, tone-signalling push button capability. The Adapter TA-977 ( )/PT enables the Switchboard SB-22( )/PT to be used with the Switchboard, Telephone Automatic SB-3614/TT. Operation power is obtained from a 9.0 volt nominal alkaline battery, military type BA-3090/U.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A, level B or Commercial preservation and packing (See Section 5).

c. When the rough handling test is required (See 4.12).

d. When First Article samples are required (See 4.3.1).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to materials specifications or preparing activity for recommended disposability methods.

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6.4 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (See 4.5.3.3).

6.5 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TA-977W/PT. The contractor should apply nomenclature in accordance with the applicable clause in the contract.

6.6 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Location of Operation Inspection. It is desirable that the operational inspection (4.9.7) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Definitions.

6.8.1 Branching. Branching is a connected arrangement of filaments (hyphol) formed by shoots or secondary items growing from the main stem or filament (hypha).

6.8.2 Growth Colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.8.3 Microbial Growth. Microbial growth is the growth of very minute organisms. Such organisms, when present in large numbers may provide a colony visible to the naked eye.

6.8.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.8.5 Tubular Germination. Tubular germination is partial growth by the production of hyphaes, which are tubular shaped fungae filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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6.8.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

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Army - CR)

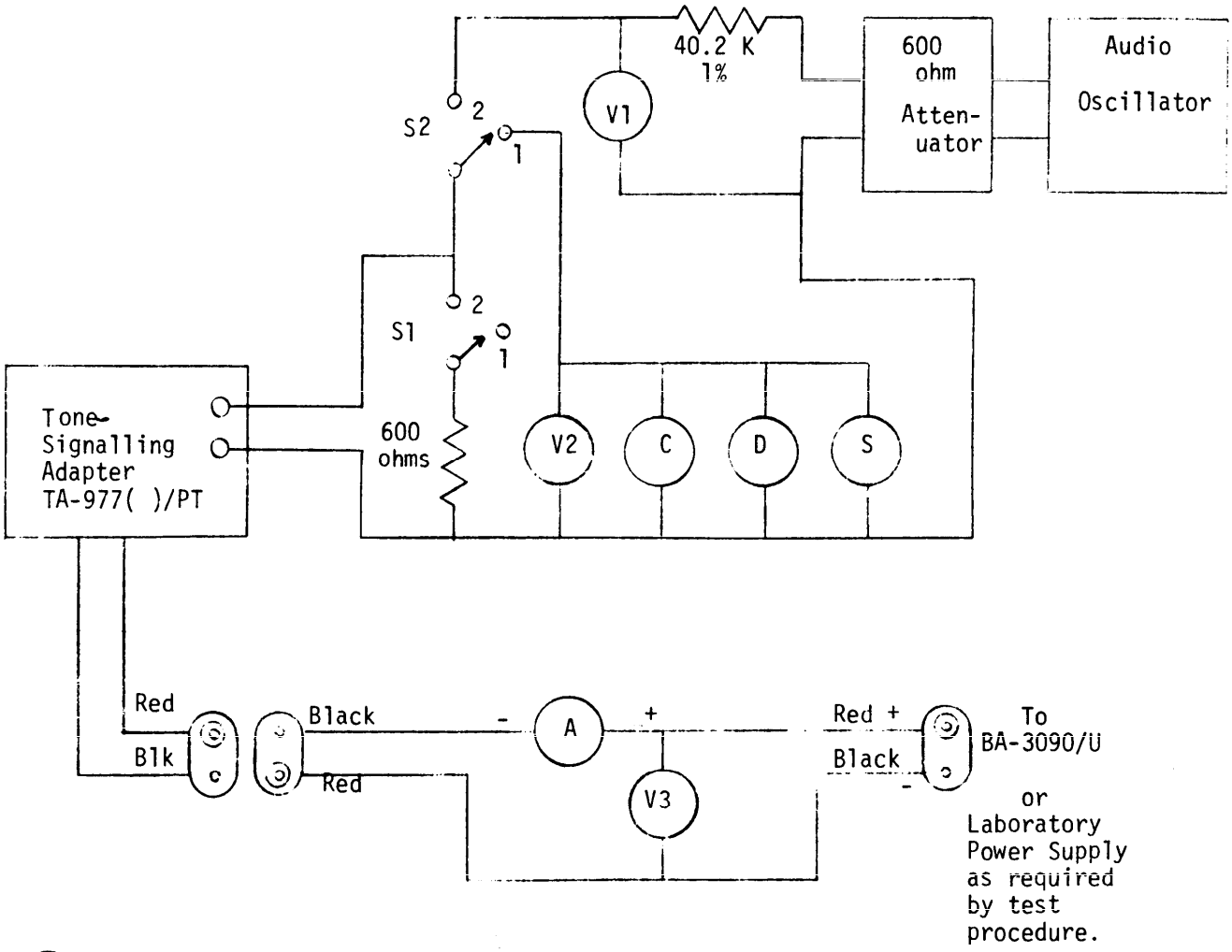
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ADAPTER TONE-SIGNALLING
TA-977( )/PT OUTPUT TEST SETUP



(A) D.C. Milliammeter

(V1) H.P. 400D, AC VTVM; or equivalent

(V2) H.P. 400D, AC VTVM; or equivalent

(C) Digital Counter

(D) Distortion Analyzer

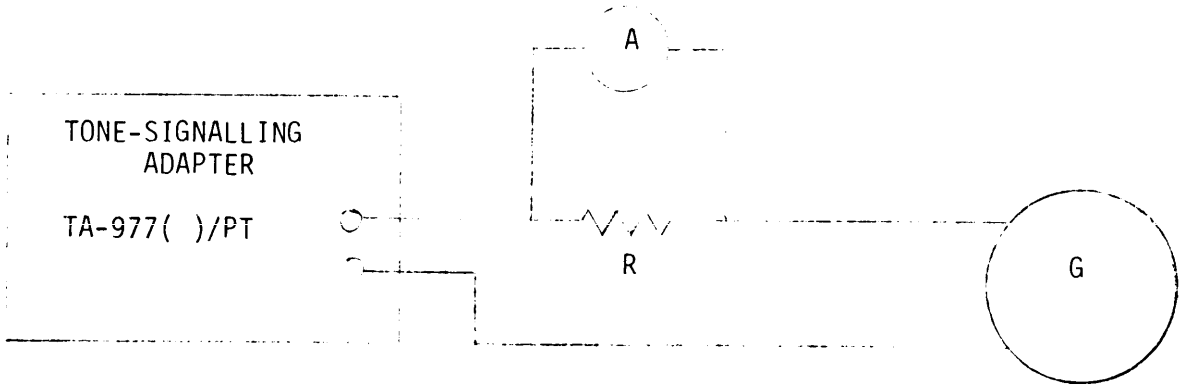
(S) Oscilloscope

(V3) D.C. Voltmeter

NOTE: During environmental testing, the power source and test equipment shall be external to the test chamber.

Figure 1

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R = 100 ohms, 5%, 1 watt, carbon-composition

A = AC VTVM H. P. 400H; or equivalent

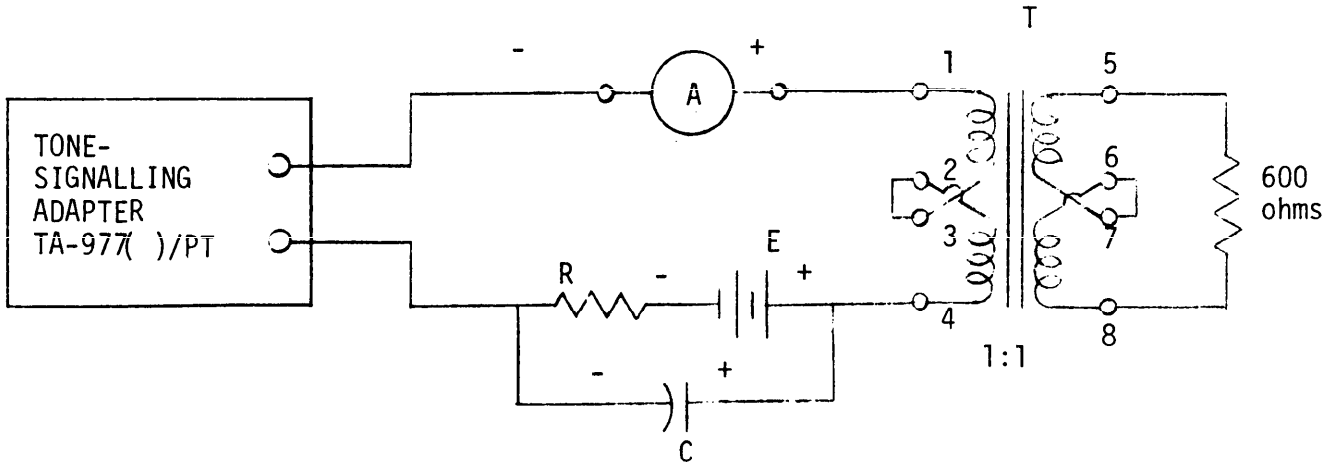
G = Generator, Ringing, Hand G-42A/PT

NOTE: BA-3090/U Battery shall be in place after test. (See 4.9.4).

Figure 2

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ADAPTER TONE-SIGNALLING TA-977( )/PT
ISOLATION TEST SETUP



R = 500 ohms, 5 Watts

C = 100 MFD, 50 W. V.D.C.

T = UTC 0-30 or equivalent

E = 24 Volts

(A) = DC Milliammeter; H.P. 412 A (.001 milliamp. lowest scale) or equivalent.

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Figure 3

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